

Chirag Deepak Shinde

Chicago, IL | (464) 300-6978 | cshin29@uic.edu | [linkedin.com/in/chirag-d-shinde](https://www.linkedin.com/in/chirag-d-shinde)

Profile

Data Scientist with hands-on experience in statistical modeling, machine learning, and end-to-end ML pipeline development. Built predictive models achieving 85% forecast accuracy and applied NLP and LLM workflows to extract insights from large unstructured datasets. Skilled in Python, PySpark, SQL, and Azure data-engineering tools, with familiarity in deep learning frameworks. Seeking to leverage data science and engineering expertise to drive data-driven decision making and improve business outcomes.

Skills

- **Machine Learning & AI:** Scikit-learn, NLP, RoBERTa, SentenceTransformers, LLM Workflows (Gemini/OpenAI APIs), RAG Architectures, Vector Databases (FAISS, PostgreSQL), Multimodal AI
- **Statistics & Analytics:** Statistical Modeling, Time-Series Forecasting, Regression (OLS, Negative Binomial), ANOVA, A/B Testing, EDA, Hypothesis Testing
- **Programming & Data:** Python (Pandas, NumPy, Scikit-learn), SQL, PySpark, Apache Spark, Jupyter Notebooks, Distributed Processing, CSV, JSON
- **Data Engineering & Cloud:** Azure Data Factory, Databricks, Snowflake, Azure Synapse, Azure ADLS Gen2, Delta Lake, ETL/ELT, Data Modeling
- **Visualization & DevOps:** Power BI (DAX), Streamlit, MLflow, Git, Docker, CI/CD, Dagster

Professional Experience

University of Illinois at Chicago

Sep 2025 - May 2026

Graduate Research Assistant

Chicago, IL

Project: Board Director Mortality & Stock Price Impact Study

- Built an ML-ready research dataset by engineering automated ETL pipelines using Azure Data Factory and Gemini Flash LLM, extracting unstructured JSON and CSV web data into Azure Data Lake - reducing manual data collection effort by 60%.
- Improved feature quality for predictive modeling by designing automated PySpark validation and transformation workflows on Azure Databricks, applying Unity Catalog governance to ensure schema consistency and data integrity.
- Accelerated model iteration cycles by profiling datasets, mapping entity relationships, and optimizing a normalized data model - cutting SQL query runtime and enabling direct integration with Azure Synapse and Power BI.
- Increased research outreach scalability by 50% by automating alumni and startup campaign pipelines using Power Automate integrated with Azure SQL.

M.S Engineers

May 2023 - Jul 2024

ML Engineer / Data Scientist

Pune, IND

Project: Sales Forecasting & Analytics Data Platform

- Increased sales forecast accuracy to 85% by building a time-series ingestion and transformation pipeline using Azure Event Hubs and batch processing to deliver clean, feature-engineered data to Python ML models - improving forecast precision by 15%.
- Improved ML model reliability by implementing a Spark SQL data quality framework with null checks, schema drift detection, statistical threshold validation, and automated A/B comparisons on live data streams.
- Reduced feature engineering overhead by 30% by building medallion-style ETL pipelines with PySpark and Snowpark in Snowflake, standardizing data transformations across Bronze, Silver, and Gold layers.
- Supported BI and model reporting by designing Star Schema data models and structured data marts, enabling scalable KPI analysis and reporting pipelines.
- Automated model retraining workflows and tracked experiments using MLflow by building a CI/CD DataOps pipeline with Docker, GitHub Actions, and Git - reducing deployment overhead by 50%.

Projects

RAG-based Q&A System with Vector DB

Aug 2025 - Dec 2025

- Reduced document search time by 70% by building a PySpark-based ingestion pipeline with SentenceTransformer embeddings and FAISS (HNSW/IVF) vector indexing for citation-grounded semantic search.
- Improved retrieval precision by designing structured metadata schemas and hybrid filtering in PostgreSQL alongside high-dimensional vector search.
- Delivered a production-ready AI application integrating a Gemini LLM backend, Dagster orchestration, and a Streamlit frontend for real-time document Q&A.

YouTube Mental Health Recovery Analysis

Jan 2025 - May 2025

- Extracted 80+ structured features using an ETL pipeline with YouTube API, PySpark, Dagster, and RoBERTa/Gemini LLM workflows for NLP-based sentiment and engagement analysis.
- Identified statistically significant engagement drivers by applying Negative Binomial Regression, ANOVA, and OLS modeling to link emotional indicators with audience metrics.
- Delivered actionable insights by integrating results with Power BI dashboards and optimized Python SQL queries for multi-dimensional visualization.

Education

University of Illinois at Chicago (UIC)

Aug 2024 - May 2026

Master of Science, Computer Science

Chicago, IL

SPPU University

Aug 2019 - May 2023

Bachelor, Computer Engineering

Pune, IND

Certifications

- Databricks Certified Data Engineer Associate | Databricks | June 2026